Dynamic Offloading Based on Lyapunov Optimization for UAV-Assisted Maritime IoT-MEC Networks

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***Abstract*—With the rapid expansion of Maritime Internet of Things(M-IOT), a number of computation-intensive and latency-sensitive tasks are increasing. To cope with this challenge, the convergence of maritime Uncrewed Aerial Vehicle (UAV) and Mobile Edge computing (MEC) has become a much-anticipated solution for intelligent maritime systems, which can effectively improve the service by delivering computationally intensive tasks to UAVs for processing. In this paper, we consider a UAV-assisted M-IoT network in which task data arrive randomly. Our goal is to design an online algorithm that can minimize devices' transmission energy and UAV propulsion energy while satisfying queue stability and making decisions for each time slot without knowing the future data arrivals. We formulate the problem as a multi-phase linear programming, and proposed a low-complexity delay-aware dynamic offloading (DADO) based on Lyapunov optimization to solve the coupling effect of the control actions in consecutive time. This algorithm transforms the long-term decisions into deterministic optimization within each time slot. The algorithm can realize the trade-off between energy and delay and make adaptive decisions based on the state of the network. Finally, the experimental results show that, comparing with other algorithms, our proposed algorithm effectively reduces the energy consumption and delay while also ensuring queue's long-term stability.**

***Index Terms*—Mobile edge computing (MEC), uncrewed aerial vehicle (UAV), Lyapunov optimization, energy efficient.**

I. Introduction

THE rapid development of the Industrial Internet of Things (IoT) has driven the rise of various smart applications in recent years, including but not limited to autonomous driving, virtual reality, face recognition, online entertainment, and smart oceans, and so on [1]. Due to the rapid development of the maritime financial market, the concept of the Internet of Things (IoT) extends to maritime networks, and the Maritime Internet of Things (M-IOT) has become a representative IoT application by virtue of its own enormous economic potential. In addition to traditional marine transportation and fishing operations, activities including seabed resource exploitation, maritime military, and marine environment monitoring are becoming more significant [2].

In this context, MEC is widely used in maritime IoT. MEC "sinks" services and functions that are originally located in the cloud center to the edge, providing strong computing power for maritime terminals with limited resources [3]. MEC servers can efficiently reduce transmission latency and energy consumption since they are closer to the terminal devices and have a shorter transmission distance than cloud centers [4]. However, traditional MEC base stations suffer from fixed locations, high deployment costs and inadequate infrastructure. To solve these problems, we choose to apply UAVs to MEC to improve the performance of maritime communication networks by deploying edge servers on UAVs [5]. The UAVs are used as airborne MEC servers to extend the service and complement the coverage of the terrestrial system.

However, the development of maritime communications still lags behind that of land-based communications due to environmental and construction difficulties. Compared with land-based communications, maritime networks have a complex and unstable environment, such as blockage due to sea clutter, long communication distances and unstable channels [6]. Currently, maritime communication networks are mainly composed of maritime radio systems, shore-based cellular networks and satellite communications. The current maritime communication networks based on high frequency (HF) and very high frequency (VHF) have low capacity and cannot be used to transmit large amounts of data services [7]. Due to the limited coverage range of coastal infrastructure, the transmission distance is restricted to approximately 100 kilometers, serving only a portion of the users. Although satellite terminals overcome the coverage problem, they possess extremely high service costs. With the diversification of marine activities and the increase in people's needs, the number of computationally intensive tasks has increased, and it is difficult for devices to handle massive computationally intensive tasks in an energy-efficient and low-latency manner. There is still a need to investigate cost-effective and high-speed communication structures for maritime networks. Therefore, research on maritime communication networks is of great importance. Recently, a number of researchers have made progress in the integration of UAVs and MEC. Zeng Yong in

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[8] considered a flight-hover-communication design for UAVs to optimize the hover locations and durations as well as the flight trajectories through the use of traveler's problem with neighborhoods (TSPN) and convex optimization techniques. The authors in [9] proposed an optimal flight trajectory algorithm for UAVs based on traffic situational awareness and an energy efficient autonomous deployment strategy based on deep reinforcement learning to optimize the trajectory and energy consumption of UAVs. Author Qian Liping in [10] combines the non-orthogonal multiple access technique with maritime MEC to minimize the total energy consumption by jointly optimizing the offload load, transmit power, computational resource allocation of USVs, and UAV trajectories that satisfy the USV latency requirements. Zhuang et al. in [11] proposed a novel adaptive network routing method based on deep reinforcement learning and designed a blockchain-based protocol to self-adapt to changing network traffic, achieving excellent results in terms of squeeze size, throughput, and latency. Author Liu Ying in [6] builds a two-tier UAV maritime communication network that combines Deep Q-Network (DQN) and Deep Deterministic Policy Gradient (DDPG) algorithms to optimize the trajectories of T-UAVs and the configuration of VMs, aiming to solve the problem of minimizing the latency of both communication and computation in this maritime UAV clustered mobile edge computing network. In [12], the authors proposed the FRES framework, which effectively addresses the resource scheduling problem in IRS and UAV-assisted MEC systems by combining a multitask agent, a progressive scheduler, and a light taboo search. In addition to this, Li et al. [13] introduced a large AI model-based edge intelligence offloading framework (LAMBO), which resolves heterogeneous constraints and dynamic environmental adaptability issues through input embedding and an asymmetric encoder-decoder architecture, significantly enhancing generalization capabilities and decision-making efficiency in multi-task scenarios. Author Yu Zhe in [14] further investigates the collaboration between UAVs and multiple terrestrial edge clouds to minimize the weighted sum of UAV energy consumption and service latency for all users. However, the above studies usually assume that the task arrival rate and channel state are known or can be predicted from historical data. And in resource-constrained scenarios, it may be difficult for devices to support high-complexity models. In real scenarios, task arrivals often change dynamically, and the wireless channel quality is interfered by multiple factors. Therefore, these parameters may be difficult to be predicted accurately.

To address this problem, some researchers have also used a variety of stochastic optimization strategies with low complexity, such as the Liapunov optimization strategy, which are able to achieve real-time online decision making without any a priori information [15]. Author Mukherjee in [16] utilizes "Lyapunov drift + penalty function" to maximize the amount of tasks accomplished within the task duration. Yang Zheyuan considered a UAV-MEC system with energy harvesting devices in [17], where a disturbance Lyapunov optimization-based online algorithm was designed to jointly optimize the energy and task processing rates of the UAV. This algorithm ensures long-term stability of the data queue while meeting the system requirements. Du Jianbo designed an iterative algorithm based on continuous relaxation and Lagrangian dual decomposition in [18], which jointly allocates wireless radio resources and power. This algorithm flexibly balances the trade-off between cost and delay, thus reducing energy consumption. In [19], Maharjan Sabita proposed a novel paradigm of digital twin networks, establishing network topology and a stochastic task arrival model. They introduced an asynchronous participant-critic algorithm to identify the optimal offloading strategy. Zhang Tong in [20] focus on the offloading problem of IoT devices in two-layer MEC networks and propose an online algorithm that effectively reduces system energy consumption. Bi Suzhi in [21] proposed an online UAV trajectory and offloading algorithm with the aim of minimizing the energy consumption of mobile devices with energy harvesting devices.

However, existing offloading strategies based on Lyapunov optimization tend to focus only on energy consumption or latency, and are unable to optimize energy saving while effectively sensing the backlog state of the queue, and these schemes are not suitable for handling latency-sensitive and computation-intensive tasks. On the one hand, while a UAV deployed with a MEC server provides high-quality services to multiple devices, the offloading process of the tasks consumes a lot of energy, such as the energy consumed during task transmission, and the changes in the trajectory of the UAV directly affect the transmission energy consumption of different devices. On the other hand, the state of the wireless channel also directly affects the transmission energy consumption of IoT devices. When the channel conditions deteriorate, the transmission process consumes more energy. Therefore, we need to divert for the task while satisfying the user's demand, and try to choose to offload the task in the case of better channel conditions. However, in real maritime environments, network quality, channel state and task arrival are complex, dynamic and unpredictable [22]. If we simply allow tasks to offload tasks when the channel state is good, although it can reduce energy consumption, it can also lead to higher queue backlogs, which in turn can trigger task queue congestion and increase latency. Therefore, our offloading strategy needs to take into account the tradeoff between queue backlog and energy consumption at the same time, as a way to achieve long-term stability and efficiency of the system.

Our goal is to design a low-complexity on-line offloading algorithm that makes decisions within each time frame without knowledge of random channel conditions and future data arrivals, in order to achieve long-term stability of the data queue and minimize system energy consumption. Our main contributions are as follows:

1) We consider a UAV-assisted MEC system in a maritime environment, where mission data arrives randomly in consecutive time frames while considering the mobility of UAVs and devices. We formulate the system energy minimization problem as a multi-stage stochastic optimization problem, defining a virtual queue to sense the backlog level of the actual queue, thus assisting in task offloading.
2) We apply perturbed Lyapunov optimization to solve the problem of high temporal coupling of control decisions, so that the device only relies on the current state to make

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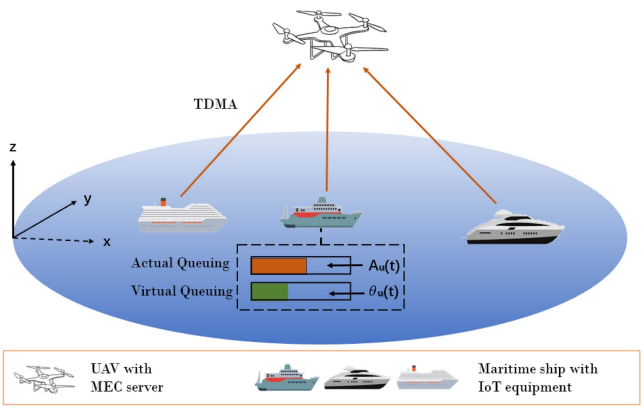

Fig. 1. Maritime mobile edge computing system model.

decisions without the need to anticipate future task arrivals and channel conditions. Meanwhile, by introducing virtual queues, we transform the task duration constraint into a queue stability control problem. Our proposed DADO algorithm with low complexity achieves the joint optimization of task offloading and UAV trajectory, and the algorithm has high real-time and adaptability.

3) We also prove the asymptotic optimality of the proposed algorithm. It is shown that the algorithm is able to achieve a good trade-off between energy consumption and task delay. By choosing appropriate parameters, the algorithm can ensure the stability of the system while keeping the energy consumption close to the optimal solution. Simulation results show that our proposed DADO algorithm is able to better reduce energy consumption and keep the queue stable compared to other benchmark algorithms.

The remaining article is organized as follows: The system model and problem formulation are defined in Section II. The online offloading algorithm based on Lyapunov optimization is presented in Section III. The performance of DADO is analyzed theoretically in Section IV. The simulation results are discussed in Section V. Finally, we summarize and present the future work in Section VI.

II. System Model and Problem Formulation

A. System Model

As shown in Fig. 1, we consider a maritime mobile edge computing system model that contains a layer of mobile devices and a layer of UAVs flying at a fixed altitude H, each UE is equipped with a simple task processor. The UAV is equipped with a powerful MEC unit and flies at a constant altitude H, simultaneously serving all devices over the duration. Assume that a UAV serves U maritime UEs, where device $u \in \{1, 2, \cdots, U\}$, each UE possesses a processor for a simple task. We divide the duration into t sub-time slots of duration Δ. The amount of random tasks arriving at device u in the $t-$ th sub-time slot is denoted as $A_u(t) \in A(t) = \{A_1(t), A_2(t), \cdots, A_U(t)\}$, and$t \in \{1, 2, \cdots, T\}$ which obeys an independent homogeneous distribution $A_u(t)$ on time slots with mean $\mathbb{E}\{A_u(t)\} = \lambda_u$.

B. Communication Model

We use Time Division Multiple Access (TDMA) technology to offload services for transmissions between UAV and devices, assuming that each device can access only one sub-channel at a time, and each device can access different sub-channels at different time slots [23]. In practice, channel resources are limited, so it is assumed that there are $K(t)$ subchannels available at time slot t, with each subchannel possessing the same bandwidth W. Since the duration of each time slot is small enough, the positions of the UAV and the device at each time slot can be considered equivalent to their positions at the beginning of the time slot. Here, the UAV's position at time slot t is denoted as $p_n(t) = \{x_n(t), y_n(t)\}$, and the device's position is denoted as $p_u(t) = \{x_u(t), y_u(t)\}$. The channel between the UAV and the device can usually be modelled as LoS (Line-of-Sight) propagation [24], then the distance between the UAV and the device is denoted by$d_u(t) = \sqrt{\|p_n - p_u\|^2 + H^2}$.We denote the probability of LoS between the UAV and the device at moment t as a logistic function of the elevation angle, i.e:

$$P_{u,LoS}(t) = \frac{1}{1 + a\exp\left(-b\left(\varphi_u(t) - a\right)\right.}, \tag{1}$$

$$\varphi_u(t) = \frac{180}{\pi}\sin^{-1}\left(\frac{H}{d_u(t)}\right), \tag{2}$$

where $\varphi_u(t)$ is the elevation angle, $a$and b are environmentally relevant parameter, we assume that the variation of elevation angle within a time slot is negligible because the horizontal motion of the UAV is relatively small compared to the altitude. The wireless transmission between the UE and the UAV suffers from path loss and small-scale fading [25]. We denote the channel gain between the UAV and the device at moment t:

$$h_u(t) = \begin{cases} \sqrt{h_0 d_u^{-\tilde{\xi}}(t)} \cdot \tilde{h}_u(t), & LoS\ link \\ \sqrt{\varepsilon h_0 d_u^{-\tilde{\xi}}(t)} \cdot \tilde{h}_u(t), & NLoS\ link, \end{cases} \tag{3}$$

where h_0 is the path loss at the reference distance ($d_0 = 1$ m), $\tilde{\xi}$ is the path loss exponent, ε is the additional fading factor due to NLoS, $\varepsilon < 1$. $\tilde{h}_u(t)$ denotes small-scale fading, which is usually a complex-valued random variable and satisfies $\mathbb{E}[|\tilde{h}_u(t)|^2] = 1$. It can be seen that $h_u(t)$ is a random variable with two different degrees of randomness. We average $h_{un}(t)$ to obtain the expected channel gain:

$$\mathbb{E}\left[|h_u(t)|^2\right] = h_0 d_u^{-\tilde{\xi}}(t)\,\hat{P}_{u,LoS}(t), \tag{4}$$

where $\hat{P}_{u,LoS}(t) = P_{u,LoS}(t) + (1 - P_{u,LoS}(t)) \cdot \varepsilon$ is the regularized LoS probability taking into account the NLoS channel fading effect. Then the transmission rate of the device at time slot t is:

$$r_u(t) = W\log_2\left(1 + \frac{P_u^{tran}|h_u(t)|^2}{N_0 \cdot \Gamma}\right), \tag{5}$$

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where P_u^{tran} is the fixed transmit power, N_0 denotes the noise power, and $\Gamma > 1$ denotes the difference in channel capacity due to the adoption of practical modulation and coding schemes.

C. Queuing Model

Considering the real situation that tasks arrive dynamically and randomly, it is possible that tasks cannot be completed within a single time slot. Hence, we establish both an actual task queue model and a virtual queue model to assist offloading for each device, in which the arriving tasks are divisible and follow the first-in first-out queuing (FIFO). The queue is represented as follows:

$$Q_u(t+1) = \max\{Q_u(t) - D_u(t), 0\} + A_u(t), \tag{6}$$

$$B_u(t+1) = \max\{B_u(t) - D_u(t), 0\} + \theta_u(t), \tag{7}$$

$$\theta_u(t) = \begin{cases} 0, Q_u(t) = 0 \\ \theta_u, Q_u(t) > 0 \end{cases}, 0 < \theta \mathrm{u} < \lambda \mathrm{u}. \tag{8}$$

$Q_u(t)$ is the actual task queue backlog (in bits) of device u at time slot t, and $B_u(t)$ is the virtual task queue backlog of device u at time slot t. $D_u(t)$denotes the amount of tasks offloaded at time slot t in bits. $\theta_u(t)$ is the parameter that controls the growth of the virtual queue. It represents the rate at which the virtual queue receives tasks in each time slot. Even if no new tasks arrive in the actual queue during a time slot, the virtual queue will still grow at a rate of θ_u. Meanwhile, the processing speed of the virtual queue is the same as that of the actual queue. It is used to indirectly reflect the system load and guide offloading decisions. A smaller θ_u makes the backlog of the virtual queue less pronounced, reducing the system's perception of task accumulation and increasing the worst-case task delay. Conversely, choosing a larger θ_u will trigger task offloading more aggressively, thereby reducing the average task queuing time but also increasing the transmission energy consumption of the device. Therefore, we typically set θ_u as a constant greater than 0 but less than λ_u. Only when the actual task queue backlog is greater than 0 does $\theta_u(t)$ take θ_u.

In order to ensure that the queue backlog is finite on the time average, we need to maintain queue stability while ensuring that tasks are completed within their deadlines.And ϵ is the upper bound. Physical meaning of data queue stability is that the average input is the same as the time-averaged throughput, and each accessed task experiences a finite queuing delay, which can be shown as:

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \sum_{u=1}^{U} \mathbb{E}[Q_u(t)] < \epsilon, \tag{9}$$

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \sum_{u=1}^{U} \mathbb{E}[B_u(t)] < \epsilon. \tag{10}$$

D. Task Processing Model

Assuming that the UE's processing capacity is limited, all of the intensive tasks in the queue must be sent to the UAV for processing. The tasks that have not been transmitted in this time slot are stored in the queue waiting for transmission, following the FIFO. Therefore, the energy consumption of UEs primarily stems from the energy expended during transmission. In each time slot, part of the tasks in the queue are offloaded to UAV. We assume that the MEC server on the UAV has sufficient computational resources.Therefore, the delay caused by task execution is neglected [26], [27]. On the other hand, since the computational results are typically much smaller in size compared to the original tasks, the communication resources required, such as bandwidth and energy consumption, are minimal. Consequently, we choose to disregard the impact of result downloading on the system's delay and energy consumption [17], [26], [28]. We assume that the offloading duration of the device in time slot t is $\alpha_u(t)$, then the amount of tasks offloaded $D_u(t)$ and the energy consumed by the device for transmission $E_u(t)$ in time slot t are described as follows:

$$D_u(t) = \alpha_u(t) \cdot r_u(t), \tag{11}$$

$$E_u(t) = \alpha_u(t) \cdot P_u^{tran}, \tag{12}$$

$$0 \le \alpha_u(t) \le \Delta, \sum_{u=1}^{U} \alpha_u(t) \le K(t) \cdot \Delta, \tag{13}$$

$$0 \le D_u(t) \le \min\{Q_u(t), B_u(t)\}. \tag{14}$$

In each time slot t, $\alpha_u(t)$ must not exceed the duration of the time slot. Since different devices can access the same sub-channel at different time slot, the total offloading duration allocated to all devices must not exceed the total communication duration of all sub-channels, i.e., $K(t) \cdot \Delta$ [29]. To ensure the proper operation of the queue without encountering negative values, we simultaneously restrict the offloading time to not exceed the actual capacity available for offloading in the queue.

E. Energy Consumption Model

Due to the fact that in practical scenarios, the energy consumption of UAVs primarily stems from propulsion. Therefore, we choose to ignore the computational energy consumption, which is consistent with the approach in [17], [30]. We adopt a rotorcraft UAV model and represent the energy consumption of the UAV in time slot t as:

$$E_{uav}(t) = \left(c_1 \left(1 + \frac{3v^2(t)}{U_{tip}^2} \right) + c_2 v^3(t) + c_3 \left(\sqrt{1 + \frac{v^4(t)}{4}} - \frac{v^2(t)}{2} \right)^{\frac{1}{2}} \right) \cdot \Delta, \tag{15}$$

$$v(t) = \frac{\|p_n(t+1) - p_n(t)\|}{\Delta}, \tag{16}$$

$$0 \le v(t) \le v_{\max}, \tag{17}$$

where c_1, c_2 and c_3 are parameters related to aircraft weight, wing area, air density, etc., and U_{tip} is the tip speed of the rotor blades [31]. The UAV's flight speed v must not exceed the maximum flight speed $v_{\max}$. The energy consumption of the system in time

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slot t is expressed as:

$$E(t) = \sum_{u=1}^{U} E_u(t) + E_{uav}(t). \tag{18}$$

We aim to minimize system energy consumption by optimizing the offload duration of the device and the flight trajectory of the UAV. We consider a long-term dynamic maritime MEC environment, which includes stochasticity in the number of available channels, stochasticity in the amount of mission data arriving, and mobility of UAVs and devices. The problem is modeled as follows:

$$\begin{aligned} P1: \quad & \min_{\alpha_u(t), p_n(t)} \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[E(t)] \\ \text{s.t.} \quad & (9), (10), (13), (14), \text{ and } (17), \end{aligned} \tag{19}$$

where (9) and (10) denote constraints on the length of the actual queue and virtual queue. (13) and (14) represent constraints on the offloading duration. And (17) indicates constraints on the UAV's flight speed. Our algorithm encourages devices with high data queue backlog to increase the offloading duration. However, blindly increase in the offloading duration leads to an increase in the energy consumption of the device, and thus needs to satisfy the trade-off between offloading decisions and energy consumption.

In the absence of information about future task arrivals and channel resources, it is very difficult to achieve long term stability by optimizing decisions for the current time slot alone. Additionally, we need to make real-time decisions within each short time frame due to rapidly changing channel conditions. In order to avoid decision-making taking too long, the algorithm must also be sufficiently simple and fast. Therefore, in the next section, we will introduce a delay-aware dynamic offloading algorithm based on Lyapunov optimization, named DADO. Without any prior knowledge, DADO adjusts parameters to achieve system stability under long-term objectives while minimizing system energy consumption.

III. Lyapunov-Based DADO Algorithm

In this section, the original problem is first converted into a real-time decision problem by applying Lyapunov optimization. Then the problem is decomposed into an offloading decision problem and a UAV trajectory problem, and a delay-sensitive dynamic offloading algorithm (DADO) is designed to make decisions in each time slot while ensuring the long-term stability of the queue.

A. Analyzing Problems With Perturbed Lyapunov Optimization

Due to the strong coupling of the queue states, we use perturbed Lyapunov optimization to decouple the queue into a deterministic problem in each time slot and ensure the stability of the system. To jointly control the actual and virtual queues, we define $Z(t) = [Q(t), B(t)]$, and introduce a Lyapunov perturbation function $L[Z(t)]$, with a Lyapunov drift function $\Delta L[Z(t)]$, expressed as follows:

$$L[Z(t)] = \frac{1}{2}\left(\sum_{u=1}^{U} Q_u^2(t) + \sum_{u=1}^{u} B_u^2(t)\right), \tag{20}$$

$$\Delta L[Z(t)] = \mathbb{E}\{\mathrm{L}[\mathrm{Z}(\mathrm{t}+1)] - \mathrm{L}[\mathrm{Z}(\mathrm{t})] \,|\, Z(t)\}. \tag{21}$$

In order to minimize the long-term average system energy consumption while ensuring the stability of the queue, we use the drift-plus-penalty minimization method, and the upper bound of the drift plus penalty expression is minimized at each time slot t:

$$\wedge L[Z(t)] = \Delta L[Z(t)] + V\mathbb{E}\{E(t) \,|\, Z(t)\}, \tag{22}$$

where V is a constant control parameter greater than 0, aimed at balancing energy consumption with queue backlog. Next, we will derive the upper bound of the drift-plus-penalty.

Lemma 1: For any queue state and action, the drift-plus-penalty $\wedge L(Z(t))$ has an upper bound:

$$\begin{aligned} \wedge L[Z(t)] \le\ & D + V\mathbb{E}\{E(t) \,|\, Z(t)\} \\ & + \sum_{u=1}^{U} \mathbb{E}\Big\{ Q_u(t)(A_u(t) - D_u(t)) \\ & + B_u(t)(\theta_u(t) - D_u(t)) \Big\}, \end{aligned} \tag{23}$$

$$D = \frac{1}{2}\sum_{u=1}^{U}\left\{2[D_u^{\max}]^2 + [A_u^{\max}]^2 + \theta_u^{\max}\right\}. \tag{24}$$

where $D_u^{\max}$ and $A_u^{\max}$ represent the maximum task transmission capacity of the device within a time slot and the maximum load that can be generated within a time slot, respectively.

Proof: use the inequality $(\max\{a-b, 0\} + c)^2 \le a^2 + b^2 + c^2 + 2a(c-b), \forall a, b, c \ge 0$ and combining it with (6) (7) we can obtain:

$$\begin{aligned} Q_u^2(t+1) \le\ & Q_u^2(t) + D_u^2(t) + A_u^2(t) \\ & + 2Q_u(t)(A_u(t) - D_u(t)), \end{aligned} \tag{25}$$

$$\begin{aligned} B_u^2(t+1) \le\ & B_u^2(t) + D_u^2(t) + \theta_u^2(t) \\ & + 2B_u(t)(\theta_u(t) - D_u(t)). \end{aligned} \tag{26}$$

Combined with (20), we obtain:

$$\begin{aligned} & L[\mathrm{Z}(\mathrm{t}+1)] \le \frac{1}{2} \\ & \sum_{u=1}^{U}\{Q_u^2(t) + 2D_u^2(t) + A_u^2(t) + B_u^2(t) + \theta_u^2(t)\} \\ & + \sum_{u=1}^{U}\{Q_u(t)(A_u(t) - D_u(t)) + B_u(t)(\theta_u(t) - D_u(t))\}, \end{aligned} \tag{27}$$

Substituting (27) to (21):

$$\Delta L[Z(t)] \le \frac{1}{2}\sum_{u=1}^{U}\mathbb{E}\{A_u^2(t) + \theta_u^2(t) + 2D_u^2(t)\}$$

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$$+\sum_{u=1}^{U}\mathbb{E}\left\{Q_u\left(t\right)\left(A_u\left(t\right)-D_u\left(t\right)\right)\right.$$
$$\left.+\ B_u\left(t\right)\left(\theta_u\left(t\right)-D_u\left(t\right)\right)\right\} \tag{28}$$

Then substituting (28) into (22), we obtain an upper bound for the drift-plus-penalty $\wedge L(Z(t))$, i.e.,:

$$\wedge L\left(Z\left(t\right)\right)\leq D+V\,\mathbb{E}\left\{E\left(t\right)|Z\left(t\right)\right\}$$
$$+\sum_{u=1}^{U}\mathbb{E}\left\{Q_u\left(t\right)\left(A_u\left(t\right)-D_u\left(t\right)\right)\right.$$
$$\left.+\ B_u\left(t\right)\left(\theta_u\left(t\right)-D_u\left(t\right)\right)\right\} \tag{29}$$

Thus, the proof is concluded. ■

Our objective is to minimize the upper bound of the drift-plus-penalty (23) in each time slot. And we use the technique of expected minimization in each time slot, and remove the constant term from (23), transforming the objective into solving a deterministic problem for each time slot:

$$\mathcal{P}\,2:\min_{\alpha_u(t),p_n(t)}V\left[\sum_{u=1}^{U}E_u\left(t\right)+E_{uav}\left(t\right)\right]$$
$$-\sum_{u=1}^{U}\left\{\left(Q_u\left(t\right)+B_u\left(t\right)\right)\cdot D_u\left(t\right)\right\}$$
$$\text{s.t}\ \ (9),(10),(13),(14),(17). \tag{30}$$

We can intuitively observe that $\mathcal{P}$ 2 is only relevant to the current system state and decisions (e.g., current virtual and real queue situation, arriving tasks, UAV position, etc.), satisfying the conditions for online dynamic decision-making. However, due to the coupling of control decisions, $\mathcal{P}$ 2 represents a non-convex problem. Therefore, we propose an efficient algorithm to obtain the control decisions for each time slot.

In the follow-up to this section, we further divide the problem into two parts [32], respectively: 1) offloading decision α_u. 2) UAV trajectory control p_n. We will continue to derive the solutions for each time slot problem.

B. Algorithm for Solving $\mathcal{P}$ 2

1) Solving the Offloading decision α_u: We begin by assuming that $p_n(t+1)=p_n(t)$ and removing extraneous variables, transforming the optimization problem into:

$$\mathcal{P}\,3.1:\ \min_{\alpha_u(t)}\sum_{u=1}^{U}\left[VE_u\left(t\right)-\left(Q_u\left(t\right)+B_u\left(t\right)\right)\cdot D_u\left(t\right)\right]$$
$$\text{s.t}\ (9),(10),(13),(14), \tag{31}$$

After rearranging (31), we obtain:

$$\max_{\alpha_u(t)}\sum_{u=1}^{U}\left[r_u\left(t\right)\cdot\left(Q_u\left(t\right)+B_u\left(t\right)\right)-VP_u^{tran}\right]\cdot\alpha_u\left(t\right). \tag{32}$$

At this point, we can view the offloading decision problem as a knapsack problem. We can consider $\alpha_u(t)$ as the items to be placed into the knapsack, $W_u(t)=r_u(t)\cdot(Q_u(t)+B_u(t))-VP_u^{tran}$ is the value of the items, and the available offloading time $K(t)\cdot\Delta$ as the capacity of the knapsack. To maximize the total value of the items placed into the knapsack, we greedily place high-value items into the knapsack until it is full or the value becomes negative. When the value is positive, we can obtain the offloading decision as $\alpha_{\mathrm{u}}(t)=\min\{\alpha_u^{\max},K(t)\cdot\Delta-L\}$. When the number of available channels increases, the capacity of the backpack increases accordingly, leading the algorithm to trigger task offloading more frequently. Conversely, when the communication conditions deteriorate and available bandwidth resources decrease, the knapsack capacity is reduced, resulting in increased queue backlog.

2) Solving the UAV Trajectory p_n: After obtaining the optimal offloading decision α_u^*, we can further solve the trajectory

Algorithm 1: Dynamic Offloading Algorithm DADO.

Initialization: queue backlog $Q(1)\leftarrow 0$, virtual queue backlog $B(1)\leftarrow 0$, UE positions p_u, UAV position p_n;
for t = 1 to T **do**
  Obtain the amount of tasks arrived at each device $\{A_u(t)\}_{u=1}^{U}$, and add newly arrived tasks to the queue;
  Obtain offloading decisions $\{\alpha_u^*(t)\}_{u=1}^{U}$ according to Algorithm 2;
  for each UE u **do**
    Calculate transmission speed $r_u(t)$ according to (5);
    Calculate offloaded data and execute offloading decision $\alpha_u^*(t)$ according to (11);
    Update queue backlog $Q_u(t+1)$ and $B_u(t+1)$ according to (6) and (7);
  end for
  Update UAV's position $p_n(t+1)$ by Algorithm 3;
  Update device positions $p_u(t+1)$;
end for

Algorithm 2: Determine Offloading Decision $\alpha_u(t)$.

Input: Current queue backlog $Q_u(t)$ and $B_u(t)$, control parameter V;
Output: Offloading decision α_u(t) for each device;
Initialization: offloading decision α(t) $\leftarrow$ 0,knapsack capacity $L\leftarrow 0$;
Update the knapsack capacity $K(t)\cdot\Delta$;
Update the value W for each device and arrange them in descending order to obtain W';
for each device u **do**
  Update remaining knapsack capacity$L_r=K(t)\cdot\Delta-L$;
  if $L_r>0$ **do**
    find the index of u in W';
    if $W_u\geq 0$ **do**
      $\alpha_{\mathrm{u}}(t)=\min\{\alpha_u^{\max},K(t)\cdot\Delta-L\}$;
      Update remaining knapsack capacity L_r;
    end if
  end if
end for

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control problem for UAV. For ease of representation, we remove the causality constraints and obtain:

$$\mathcal{P}3.2: \min_{p'_n} V\left(\sum_{u=1}^{U} P_u^{tran}\cdot\alpha_u^* + E_{uav}\right) - \sum_{u=1}^{U}\left[(Q_u+B_u)\cdot\alpha_u^*\cdot r_{u\{p'_n\}}\right] \quad \text{s.t (17)}, \tag{33}$$

where α_u^* is the optimal offloading decision obtained from $\mathcal{P}$ 3.1, p'_n is the optimal position of the UAV that we aim to solve, E_{uav} is obtained from (15).

We observe that $\mathcal{P}3.1$ is a non-convex function with respect to p'_n. We choose to use the Successive Convex Approximation (SCA) to solve this problem. Firstly, we introduce a relaxation variable y and auxiliary variables η_u:

$$\frac{1}{y^2} \le y^2 + v^2, \tag{34}$$

$$\eta_u \le W\log_2\left(1+\frac{P_u^{tran}\cdot h_0}{N_0\left(H^2+\|p'_n-p_u\|^2\right)}\right). \tag{35}$$

Next, we will prove that the right-hand side of (34) and (35) both have a concave lower bound.

Proposition 1: Given the local optimal point p_n^l for the $l-th$ iteration, the transmission speed of the device has a concave lower bound.

$$r^l_{u_{\{P'_n\}}} \triangleq W\log_2\left(1+\frac{P_u^{tran}\cdot h_0}{N_0\left(H^2+\|p_n^l-p_n\|^2\right)}\right) - \beta_u\left(\|p'_n-p_u\|^2-\left\|p_n^l-p_u\right\|^2\right), \tag{36}$$

where $\beta_u = \frac{W\log_2 e\cdot P_u^{tran}\ h_0}{[P_u^{tran}\ h_0+N_0(H^2+\|p_n^l-p_u\|^2)](H^2+\|p_n^l-p_u\|^2)}$.

Proof: We know that $f(z)=\log_2[1+a/(b+c)]$, where $a,b>0, z\ge 0$, and $f(z)$ is a convex function with respect to z. According to the definition of derivatives, when $f(z)$ is differentiable at z_0, it holds $f(z)\ge f(z_0)+f'(z_0)(z-z_0)$. By substituting $a=P_u^{tran}h_0/N_0$,$b=H^2$, $z_0=\|p_n^l-p_u\|^2$, $z=\|p'_n-p_u\|^2$ into the Taylor expansion, we obtain the lower bound of the transmission speed. ■

Proposition 2: Given $y^l=\left(\sqrt{1+\frac{\|p_n^l-p_n\|^4}{4\Delta^4}}-\frac{\|p_n^l-p_n\|}{2\Delta^2}\right)^{\frac{1}{2}}$, the right-hand side of (34) has a globally concave lower bound $Y^l_{\{P'_n,y\}}$:

$$Y^l_{\{p'_n,y\}} \triangleq \left(y^l\right)^2+\frac{\left\|p_n^l-p_n\right\|^2}{\Delta^2}+2y^l\left(y-y^l\right) + \frac{2\left\|p_n^l-p_n\right\|}{\Delta^2}\left(\|p'_n-p_u\|-\left\|p_n^l-p_u\right\|\right). \tag{37}$$

Proof: We perform a first-order Taylor expansion on the right-hand side of the equation to obtain $y^2+v^2\le y_0^2+v_0^2+2y_0(y-y_0)+2v_0(v-v_0)$. Substituting $v=\|p'_n-p_n\|/\Delta$ into it, we can obtain (37). ■

Algorithm 3: Determine UAV Trajectory p_n.

1: **Input:** UAV's position p_n, optimal offloading decisions α_u^* obtained from Algorithm 2;
2: **Output:** Next position of the UAV p'_n;
3: **Initialization:** Iteration threshold $\sigma=0.01$; $p_n^l=p_n$;
4: **repeat**
5: Acquire the value of y^l;
6: Solve (38) using CVX to obtain the optimal solutionp_n^* and the optimal function value M^l;
7: Assign the value of p_n^* to p_n^{l+1}; update the lower bound $Y^{l+1}_{\{P'_n,y\}}$ and $r^{l+1}_{u_{\{P'_n\}}}$;
8: $l=l+1$;
9: **until** $|M^l-M^{l-1}|<\sigma$
10: Return the optimal solution p_n^* for the next time step position of the UAV.

By using the concave lower bounds (36) and (37) to simplify the objective, the trajectory optimization in $\mathcal{P}3.2$ at the $l-th$ iteration becomes the following:

$$\mathcal{P}3.3: \min_{p'_n,y,\eta_u} V\left(\sum_{u=1}^{U} P_u^{tran}\cdot\alpha_u^* + \left(c_1\left(1+\frac{3v^2(t)}{U_{tip}^2}\right)+c_2v^3(t)+c_3y\right)\cdot\Delta\right) - \sum_{u=1}^{U}\left[(Q_u+B_u)\cdot\alpha_u^*\cdot\eta_u\right], \tag{38}$$

s.t (17),

$$\eta_u \le r^l_{u_{\{P'_n\}}}, \tag{39}$$

$$\frac{1}{y^2} \le Y^l_{\{P'_n,y\}}. \tag{40}$$

$\mathcal{P}3.3$ is a convex optimization problem, which can be solved using the optimization toolbox CVX in MATLAB [33]. After obtaining the optimal solution p_n^*, the values of $r^l_{u_{\{P'_n\}}}$ and $Y^l_{\{P'_n,y\}}$ are updated. p_n^* is considered as the new local point p_n^{l+1}, then the next iteration begins. The iteration process repeats until the absolute difference between two consecutive function values is less than the solving accuracy σ. The algorithm's workflow is illustrated in Algorithm 3 below.

IV. Algorithm Performance Theoretical Analysis

In this section, we will theoretically analyze the proposed algorithm to find out the performance gap between the optimal solution of the original problem $\mathcal{P}1$ and the optimal solution of our algorithm $\mathcal{P}2$ and verify its effectiveness. We will demonstrate the trade-off between energy consumption and latency of the algorithm through mathematical derivation and further analyze the worst-case queue of this system. Finally we will also analyze the complexity of the proposed algorithm.

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A. Optimality Gap Analysis

Lemma 2: If the original problem $\mathcal{P}1$ has a solution, then there must exist an optimal β-only policy that satisfies the following in each time slot:

$$\begin{cases} E\left\{E^{\beta}(t)\right\} = E^{*}(\lambda), \\ E\left\{A_u(t)+\kappa\right\} \leq E\left\{r_u(t)\,\alpha_u^{\beta}(t)\right\}, \end{cases} \tag{41}$$

where $E^*(\lambda)$ denotes the optimal energy consumption when the task arrival rate is λ. The proof process is similar to the method in [34]. By using Theorem 3, we further derive Theorem 1.

Theorem 1: By adopting the DADO algorithm, the average energy consumption of the system satisfies:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left\{E(t)\right\} \leq \frac{D}{V} + E^{*}. \tag{42}$$

Proof: We assume that the data arrival rate is $\lambda+\kappa$, which strictly lies within the network capacity region, where κ is a positive value. Substituting $\lambda+\kappa$ into (41), we obtain:

$$\begin{cases} E\left\{E^{\beta}(t)\right\} = E^{*}(\lambda+\kappa), \\ E\left\{A_u(t)+\kappa\right\} \leq E\left\{r_u(t)\,\alpha_u^{\beta}(t)\right\}, \end{cases} \tag{43}$$

where β' represents the offloading policy under the arrival rate $\lambda+\kappa$. Substituting (43) into (23), we obtain:

$$\begin{aligned} &\Delta L(Z(t)) + V\mathbb{E}\left\{E(t)\,|Z(t)\right\} \\ &\leq D + V\mathbb{E}\left\{E^{\beta}(t)\,|Z(t)\right\} \\ &+\sum_{u=1}^{U}\mathbb{E}\left\{Q_u(t)\left(A_u(t)-r_u(t)\,\alpha_u^{\beta}(t)\right)+B_u(t)\left(\theta_u(t)\right.\right. \\ &\left.\left.-r_u(t)\,\alpha_u^{\beta}(t)\right)\right\}. \end{aligned} \tag{44}$$

Given that $\mathbb{E}\{A_u(t)\}-\mathbb{E}\{r_u(t)\alpha_u^{\beta}(t)\}\leq 0$ and $\theta_u(t)\leq A_u(t)$, we can further simplify (44) to:

$$\Delta L(Z(t)) + V\mathbb{E}\left\{E(t)\,|Z(t)\right\} \leq D + V\mathbb{E}\left\{E^{\beta}(t)\,|Z(t)\right\}. \tag{45}$$

Combining (45) with (21), we further extend (45) to all time slots and obtain:

$$\begin{aligned} &\mathbb{E}\left\{\mathrm{L}(\mathrm{Z}(t)) - \mathrm{L}(\mathrm{Z}(0))\,|Z(t)\right\} + V\sum_{t=1}^{T}\mathbb{E}\left\{E(t)\right\} \leq TD \\ &\quad + TV\mathbb{E}\left\{E^{\beta}(t)\right\}. \end{aligned} \tag{46}$$

Since the initial load of the queue is zero, $\mathrm{L}(\mathrm{Z}(0))=0$ and $\mathrm{L}(\mathrm{Z}(\mathrm{T}))\geq 0$. Dividing both sides of (46) by VT simultaneously, and letting $T\to\infty$, we obtain (42). ■

Theorem 2: When the long-term average system load falls within the system's computing capacity, the time-averaged queue backlog for all devices is bounded, satisfying:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left\{Q(t)\right\} \leq \frac{D+VE^{*}}{\kappa}, \tag{47}$$

each device experiences a worst-case delay of $T_{\max}$, where: $T_{\max} < (Q_u^{\max}+B_u^{\max})/\theta_u$, $Q_u(t)\leq Q_u^{\max}$,$B_u(t)\leq B_u^{\max}$.

Proof: From (42), we have $\mathbb{E}\{A_u(t)\}-\mathbb{E}\{r_u(t)\alpha_u^{\beta}(t)\}\leq -\kappa$, we can then transform (43) as follows:

$$\begin{aligned} &\Delta L(Z(t)) + V\mathbb{E}\left\{E(t)\,|Z(t)\right\} \\ &\leq D + V\mathbb{E}\left\{E^{\beta}(t)\;|Z(t)\right\} \\ &\quad + \sum_{u=1}^{U}\mathbb{E}\left\{Q_u(t)\left(A_u(t)-r_u(t)\,\alpha_u^{\beta}(t)\right)\right\} \\ &\leq D + V\mathbb{E}\left\{E^{\beta}(t)\,|Z(t)\right\} - \kappa\sum_{u=1}^{U}Q_u(t). \end{aligned} \tag{48}$$

Taking the limit as $T\to\infty$ and rearranging (48), we obtain (47), implying that the queue backlog is bounded. According to Little's Law [34], the time-average delay of tasks is proportional to the queue backlog. So the task delay is also bounded. ■

After proving the boundedness of the queue backlog, our next goal is to find the worst-case delay for devices. Let's assume that a task arrives at time t, and after $T_{\max}$ time slots it still hasn't been completed, i.e., $\forall\tau\in\{t+1,t+2,\ldots,t+T_{\max}\}, Q_u(\tau)>0$. Since the queue follows the first-in-first-out principle, if a task is not completed by time $t+T_{\max}$, it must hold that $Q_u(t+1)>\sum_{\tau=t+1}^{t+T_{\max}}D_u(\tau)$.

From (7), we can derive: $B_u(\tau+1)-B_u(\tau)\geq\theta_u-D_u(\tau)$

Summing up this inequality from $t+1$ to $t+T_{\max}$, we obtain:

$$B_u(\tau+1+T_{\max})-B_u(\tau+1)\geq T_{\max}\theta_u-\sum_{\tau=t+1}^{t+T_{\max}}D_u(\tau), \tag{49}$$

where $B_u(\tau+1+T_{\max})-B_u(\tau+1)\leq B_u^{\max}$. After rearranging, we obtain:

$$\sum_{\tau=t+1}^{t+T_{\max}}D_u(t)\geq T_{\max}\theta_u-D_u^{\max}, \tag{50}$$

Then:

$$T_{\max}\theta_u-D_u^{\max}\leq\sum_{\tau=t+1}^{t+T_{\max}}D_u(t)<Q_u(t+1)\leq Q_u^{\max}, \tag{51}$$

Finally, we can obtain:

$$T_{\max}<\frac{Q_u^{\max}+D_u^{\max}}{\theta_u}. \tag{52}$$

The proof of the theorem is thus complete. ■

Theorem 1 and Theorem 2 demonstrate that the algorithm satisfies the energy causality constraint, indicating $[O(1/V),O(V)]$ trade-off relationship between system energy consumption and delay. The upper limit of average system energy consumption is inversely proportional to the coefficient V, while the upper limit of average queue backlog is directly proportional to V. To ensure queue stability, a smaller V should be chosen. However, blindly reducing the parameter V may lead to excessive system energy consumption. Moreover, combined with (24), we can also observe that the upper bounds of system delay and energy consumption are related to the maximum data

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TABLE I
SIMULATION PARAMETERS

Param	Meaning	Value
U	The number of UEs	20
H	The altitude of UAV	100m
v_{max}	Maximum flight speed of UAV	25m/s
t	Number of time slots	2000
Δ	Time slot duration	1s
P_u^{tran}	Transmission power of UEs	$P_u^{tran} \sim [10, 200]$ mw
N_0	Noise power	10^{-11}W
W	Communication bandwidth	1MHz
ξ	Pass loss exponent	2.3
a	Environmental parameter	10
b	Environmental parameter	0.6
ε	NLoS attenuation factor	0.2

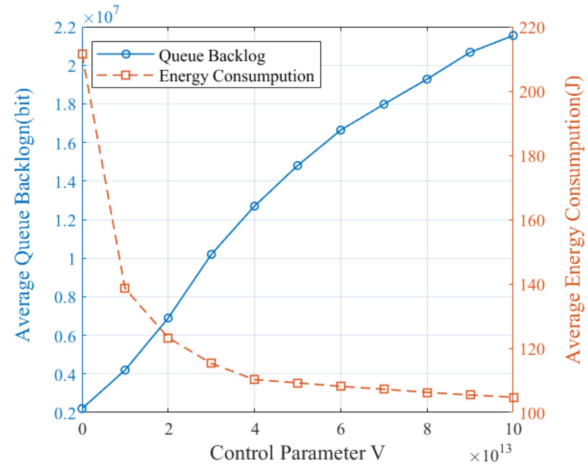

Fig. 2. Trade-off between energy consumption and delay.

arrival rate $A_u^{\max}$ and the virtual queue parameter θ_u. As the task arrival rate increases, the average queue delay also increases. Conversely, adjusting the virtual queue parameters can influence the system's average delay.

In our simulation experiments, we can choose an appropriate parameter V to ensure system stability while preventing excessive energy consumption. Additionally, we will demonstrate the system's robustness under different task arrival rates.

B. Algorithm Complexity

Next we analyze the time complexity of the DADO algorithm. The time complexity of our algorithm is expressed in terms of the number of devices U. In Algorithm 2, lines 4-5 involve calculating the knapsack capacity and item values, followed by sorting, which has a complexity of $O(U\log_2 U)$. Lines 6-13, which handle the allocation of channel resources, consist of comparison and assignment operations, also with a complexity of $O(U)$. Thus the time complexity of Algorithm 2 is $O(U\log_2 U)$. The complexity of the trajectory optimization part of Algorithm 3 is $O(U^{3.5}L\log_2(1/\sigma))$, where σ is the iteration accuracy of SCA and L is the number of iterations. Overall, the proposed algorithm has a total time complexity of $O(U^{3.5}L\log_2(1/\sigma))$

V. SIMULATION RESULTS

We conducted simulations using MATLAB to evaluate the proposed online offloading algorithm. The simulation parameters are set as shown in Table I. We consider that the UAV provides MEC service to UEs in a 1000 m × 1000 m area. The initial positions of the devices are randomly generated within the area. The initial positions of the devices are randomly generated within the area. where the initial position of the device was randomly generated in the field and moved randomly at a speed of 0 to 10 m/s. In order to ensure the reliability of the experimental results, we performed several repeated runs for each setting and averaged the results. It should be noted that the experimental parameter settings in this article are mainly used to verify the performance of the proposed algorithms and do not cover all possible application scenarios. In practice, the parameter settings can be adjusted for different MEC scenarios according to specific needs, but this will not significantly affect the core execution logic of the algorithm. Although the task arrivals and channel states are set according to a fixed statistical distribution in the experiments, the DADO algorithm does not rely on this a priori statistical information.

A. Evaluation

1) Trade-off Between Energy Consumption and Delay: According to Little's law, the average queue backlog is proportional to the delay [34], so we use the queue backlog to reflect the delay. As shown in Fig. 2, with the mean arrival rate of tasks set to $\lambda = 40000$ bit/s, and V varying from 0 to 1×10^{14}, we can observe that the average queue backlog increases continuously as V increases. Initially, energy consumption decreases rapidly, followed by a slow and gradual decrease until it stabilizes. This is because a higher value of V emphasizes energy consumption more, weakening queue control and thereby reducing energy consumption.

2) Influence of Data Arrival Rate and Control Parameter: Fig. 3(a) shows the queue backlog under different input data rates. It can be observed that as the input data rate increases, the queue backlog also increases but remains relatively stable. Fig. 3(b) illustrates the variation of queue backlog under different control parameters V. Consistent with the conclusion drawn from Fig. 2, as V increases, the system emphasizes energy consumption more, weakening queue control. However, the algorithm still converges relatively quickly. Due to the mobility of devices and the changing channel conditions, DADO continually balances energy and delay. When the queue backlog becomes too large, it adapts by selectively offloading some tasks. Conversely, when the queue backlog decreases, it chooses a more energy-efficient offloading strategy, thereby increasing the queue backlog.

3) Influence of Different Number of Devices: Fig. 4(a) and (b) respectively demonstrate the influence of varying numbers of devices on system energy consumption and average queue

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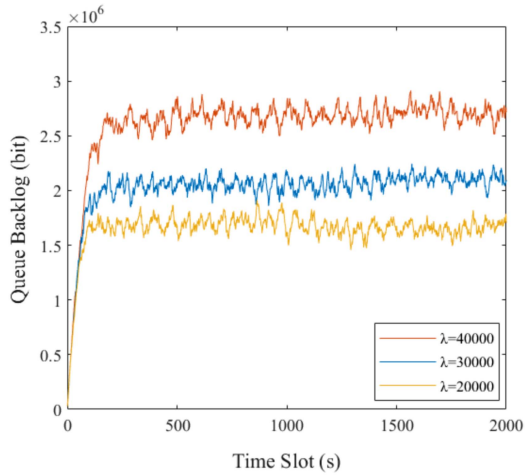
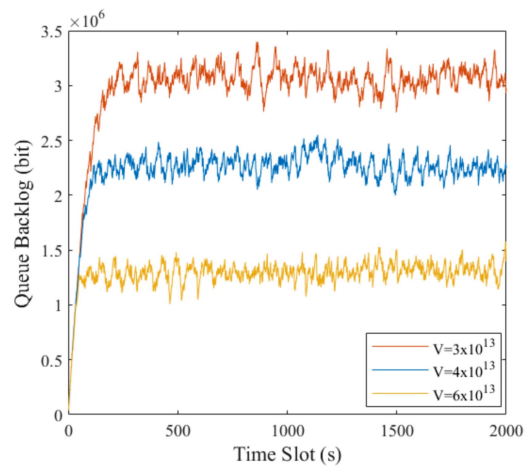


(a)



(b)

Fig. 3. (a) Influence of data arrival rate on queue backlog. (b) Influence of control parameter on queue backlog.

backlog. As observed, with an increase in the number of devices, both the average queue backlog and system energy consumption exhibit an increase. This phenomenon arises because as the number of devices escalates, the system generates a greater task load. However, due to the limited available channels, the capacity to accommodate offloaded tasks is restricted. Consequently, the system can only offload as much as possible, retaining the remaining tasks in the queue, thereby augmenting the queue backlog.

4) Impact of the Virtual Queue Parameter and the Number of Sub-Channels: As show in Fig. 5(a), with an increase in θ, the average backlog of the actual queue shows an overall decreasing trend, while the average backlog of the virtual queue increases. When the parameter θ is small, the load of the actual queue remains at a high level because the system's perception of queue backlog is weakened. As gradually increases, the backlog of actual queue decreases while the backlog of the virtual queue rises, indicating that the virtual queue is playing a greater regulatory role. Due to the trade-off between energy consumption and queue delay, the system does not blindly increase the offloading amount, leading to fluctuations in queue backlog variations.

Fig. 5(b) shows the queue backlog for different number of sub-channels. The available sub-channels are uniformly distributed in different ranges, i.e., $K(t) \sim U[0, 5]$, $U[5, 10]$, $U[10, 14]$. From the results, it can be seen that the average queue backlog increases when the number of available channels decreases. This is due to the fact that with limited channel resources, and some of the tasks cannot be offloaded in a timely manner and thus accumulate in the queue. The algorithm prioritizes the offloading of queues with higher loads, but the backlog of tasks is still high due to the overall resource constraints. Meanwhile, when there are more available channel resources, the queue fluctuates less, which is because sufficient channel resources will trigger offloading more frequently, which can empty the task queue in time to avoid large load fluctuations and thus maintain a more stable queue length.

B. Comparison of Different Algorithms

In this section, we compare the following methods:

1) *Longest Queue Length First (LQLF):* This is a baseline algorithm that prioritizes offloading tasks from the longest queue, with offloading durations allocated based on queue backlog.
2) *Energy Efficient Dynamic Offloading Algorithm (EEDOA):* This is a dynamic offloading algorithm proposed by Ying Chen et al. in 2021, which is also based on Lyapunov optimization [24].
3) *TSP-based Trajectory Optimization:* This algorithm considers the UAV trajectory optimization as a classic Traveling Salesman Problem (TSP) [35], with the objective of minimizing the total travel distance while visiting all UEs in the duration. After obtaining the fixed trajectory, we accordingly optimize the offloading strategy.

1) Comparisons Under the Same Task Arrival Rate: Fig. 6(a) shows the variation in queue backlog for four algorithms under the same task arrival rate of $\lambda = 40000$ bit/s. It can be observed that LQF and DADO have lower queue loads compared to EEDOA and TSP, with LQF exhibiting poorer stability, characterized by more pronounced fluctuations in queue load compared to the other two algorithms. The queue backlog of TSP is more stable compared to EEDOA and LQLF, while the average queue backlog is higher than that of DADO. It can be inferred that the Lyapunov-based offloading strategy outperforms other algorithms in terms of queue stability control. Thus, it can be concluded that our proposed DADO algorithm outperforms others in queue stability control and latency reduction.

Fig. 6(b) show the comparison of energy consumption among different algorithms. It can be observed that the LQLF algorithm consumes the most energy, with a large fluctuation range. This is because it always selects the longest queue for extensive offloading, leading to increased energy consumption. Due to the highly unstable queue backlog, its energy consumption also exhibits an unstable pattern. Compared to the other two algorithms, DADO demonstrates lower energy consumption and better stability. It is evident that our algorithm is effective. The TSP algorithm is inferior to the other algorithms in trajectory control and has the highest level of energy consumption.

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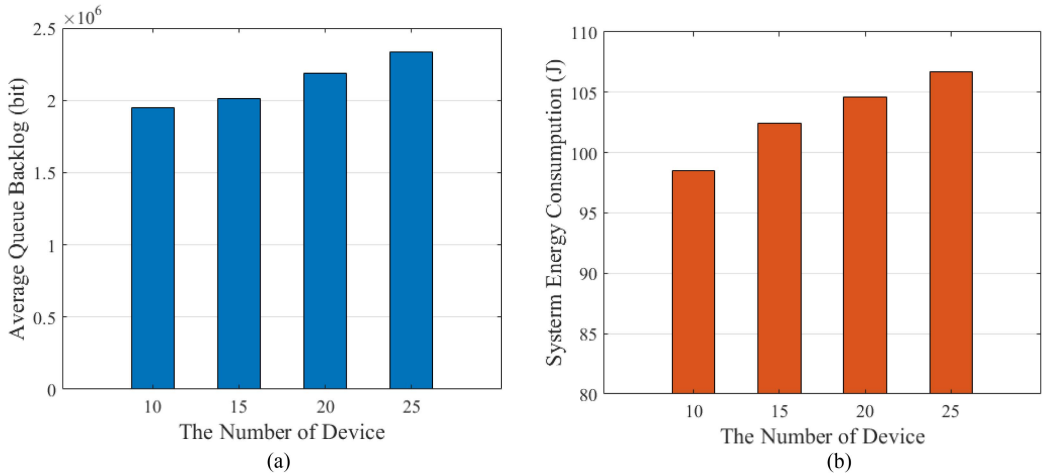

Fig. 4. (a) Impact of the number of devices on system energy consumption. (b) Impact of the number of devices on average queue backlog.

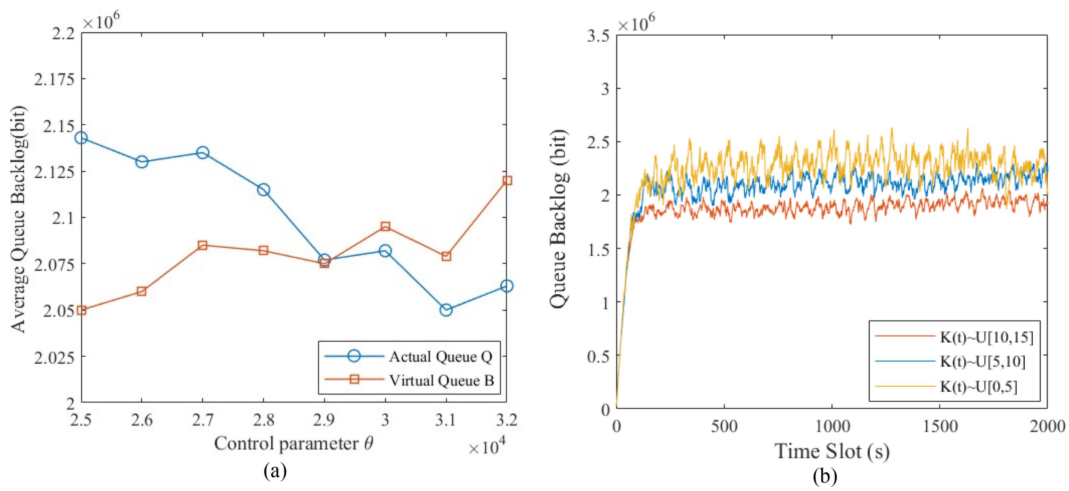

Fig. 5. (a) Impact of the control parameter θ on average queue backlog. (b) Impact of the number of sub-channels on average queue backlog.

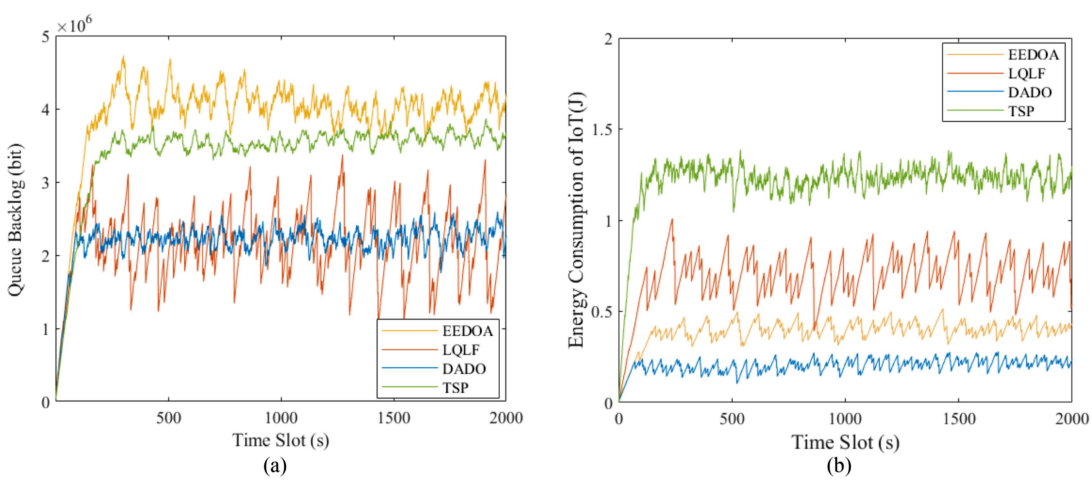

Fig. 6. (a) Comparison of queue backlog under the same arrival rate. (b) Comparison of energy consumption under the same arrival rate.

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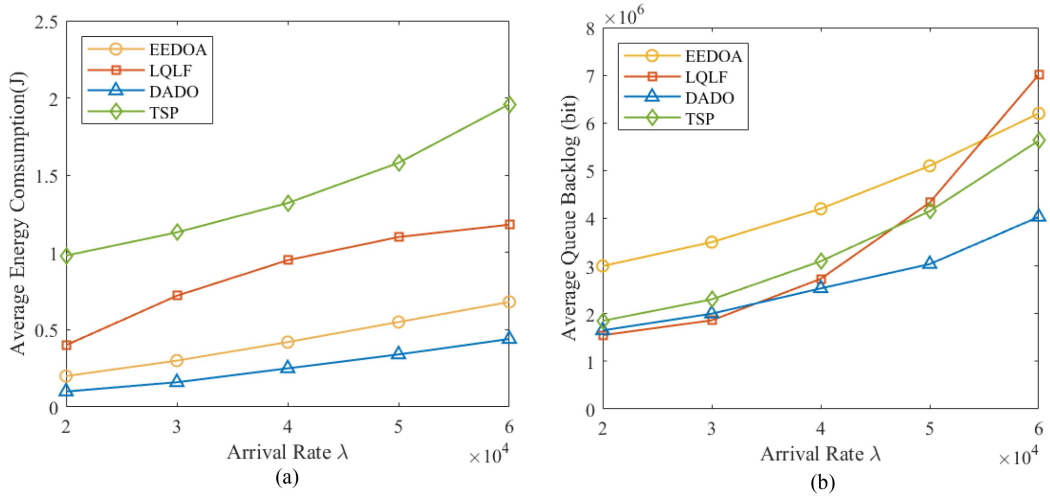

Fig. 7. (a) Average queue backlog under different arrival rate. (b) Average energy consumption under different arrival rate.

2) Comparison Under Different Task Arrival Rate.: From Fig. 7(a), it can be observed that LQLF shows a lower average queue backlog at low task arrival rate. However, as λ increases, queue backlog increases sharply, indicating that LQLF is not well-suited to handle high task arrival rate. In contrast, the growth of EEDOA and DADO is more gradual, showing the advantage of Lyapunov optimization in queue stability. Compared to other algorithms, DADO shows higher stability, with consistently lower queue backlog, especially when λ is large, making the difference even more pronounced.

Fig. 7(b) shows the average energy consumption under different λ. As the task arrival rate increases, the average energy consumption of all algorithms rises, which is reasonable since processing more tasks requires additional energy. Due to the fixed flight path of UAV in TSP, its average energy consumption remains consistently high, showing the disadvantage in energy consumption control. In summary, the proposed DADO algorithm exhibits good queue stability, low latency, and certain advantages in energy consumption control, even when λ is high.

VI. Conclusion

In this paper, we investigated the problem of minimizing energy consumption in a maritime environment under the framework of UAV-assisted mobile edge computing systems. We constructed a delay-aware queue and utilized perturbed Lyapunov optimization to transform the complex long-time-slot dynamic optimization problem into a deterministic-time-slot optimization problem. We proposed a low-complexity delay-aware dynamic offloading algorithm, named DADO, to minimize system energy consumption by decomposing the problem in each time slot into two sub-problems. Simulation results demonstrate that, by selecting appropriate parameters, the DADO algorithm can maintain queue stability in dynamic environments and achieve good performance in reducing system energy consumption.

Currently, DADO mainly focuses on a single UAV to provide computing services for a certain number of devices in a fixed area, it is the foundation for studying multi-UAV scenarios, and holds significant practical value in practical applications, such as small-scale maritime monitoring. However, in larger scale maritime IoT applications, there are limitations in the computation and coverage capabilities of a single UAV. A direct extension is to adopt a multi-UAV architecture, where different UAVs are responsible for different regional blocks as a way to increase the service scale and coverage capacity of the system. However, this regional division strategy lacks collaboration among UAVs, potentially leading to unbalanced resource utilization. In future work, we will explore multi-UAV scenarios and large-scale device deployments to further optimize UAV resource utilization. Specifically, we aim to integrate multimodal data fusion and task aggregation techniques into UAV-assisted IoT-MEC environments. Task aggregation can further enhance MEC resource utilization, for instance, by enabling UAVs to preprocess data [36], or facilitating collaborative computation among MEC servers, thereby reducing task latency and energy consumption.

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